

Derek Martin

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Professional Summary

Mechanical engineer pursuing an M.Eng. with an aerospace focus and physics background, combining mechanical design, Python control development, instrumentation, field operations, and project/business exposure. Experienced with CAD/FEA, SCADA, troubleshooting, fabrication, and multidisciplinary engineering work.

Education

Master of Engineering (M.Eng.), Mechanical Engineering | Expected June 2027

B.S. Mechanical Engineering, Aerospace Engineering Focus, Physics Minor | September 2026

Oregon State University | Corvallis, Oregon

B.S. GPA: 3.68 | Dean's List

Relevant Coursework: Space Systems Engineering, Rocket Propulsion, Applied Stress Analysis, Systems Dynamics & Control, Dynamics, Mechanics of Materials, Instrumentation

Professional Experience

Engineering Intern | Evensol Inc. - EDG-L (Landfill Gas Collection) | June 2024 - September 2024

- Supported landfill-gas production and decant operations involving compressors, pumps, valves, transmitters, gas sampling, trailer operations, preventive maintenance, troubleshooting, and equipment installation.
- Designed and modeled a fan-motor hand-crank system in Autodesk Fusion 360 and performed stress analysis to evaluate loads and potential weak points; fabricated and fitted steel components for the assembly.
- Calibrated instrumentation and assisted with condensate-pump installation, control-panel equipment, compressor maintenance, and field repairs in a safety-critical environment.
- Shadowed the CEO during project planning, feasibility analysis, budgeting, proposal development, landfill-gas project investment evaluation, and executive meetings.
- Worked under the head engineer, gaining exposure to multi-month project planning and engineering workflows; digitized and reformatted an operations quick-reference binder for plant managers.
- Supported manufacturing work involving part production workflows, tolerances, fabrication steps, and component progression from shop work to finished product.

Wildland Firefighter | USFS Sisters Ranger District | August 2022 - September 2023

- Performed direct and indirect fireline construction and operated firefighting equipment in high-risk wildfire environments.
- Maintained chainsaw certification for hazard tree and vegetation removal and used GPS/navigation tools during suppression operations.

Technical Skills

Software: Python (automation/control), Microsoft Excel, Autodesk Fusion 360, SCADA

Engineering: CAD/FEA, stress & structural analysis, instrumentation, controls, troubleshooting, fabrication

Project/Operations: Feasibility analysis, project planning, technical documentation, scheduling, data analysis

Safety: PPE, working at heights, emergency response, safety-critical field operations

Systems: Space systems fundamentals, electromechanical integration, reliability, process flow

Certifications

Emergency Medical Technician (EMT) | Certified through Impact EMS | June 9, 2024

Achievements

- Team Captain, Nordic Skiing
- Medallion awarded by Bend Police Department for integrity and discipline (2021)

Projects

- Autonomous Piano Player (Capstone, 2025-2026): Developed Python control software for an electromechanical system and supported CAD, mechanical integration, and system testing.
- Winch Lift Stand: Designed and analyzed an aluminum stand supporting a winch to raise a 150 lb motor approximately 50 ft with a high factor of safety.
- Food Stirrer Prototype: Performed stress analysis and supported iterative design for a 3D-printed liquid food stirrer with a 4-person team.